The Highlighted value in the table is the one that was changed for that model.

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay
Baseline	47.58%	64.26%	0.00%	0.00%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	1.00E-03	1.00E-04
Conv/FC Layers	49.35%	67.72%	1.77%	3.46%	17,275,384	**IN - 128 - 256 - 512**	**IN - 1024 - 100**	**IN - 512 - 20**	ReLU	YES	0.4	1.00E-03	1.00E-04
Conv/FC Layers	28.26%	55.41%	-19.32%	-8.85%	10,926,840	**IN - 128 - 256 - 512**	**IN - 512 - 100**	**IN - 256 - 20**	ReLU	YES	0.4	1.00E-03	1.00E-04
Conv/FC Layers	53.52%	67.73%	5.94%	3.47%	7,552,056	**IN - 64 - 128 - 256**	**IN - 1024 - 100**	**IN - 512 - 20**	ReLU	YES	0.4	1.00E-03	1.00E-04
Batchnorm	47.30%	61.08%	-0.28%	-3.18%	4,347,448	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	**NO**	0.4	1.00E-03	1.00E-04
Dropout	55.29%	67.73%	7.71%	3.47%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	**0.2**	1.00E-03	1.00E-04
Dropout	42.11%	60.62%	-5.47%	-3.64%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	**0.6**	1.00E-03	1.00E-04
Dropout	57.02%	68.38%	9.44%	4.12%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	**0.1**	1.00E-03	1.00E-04
Dropout	57.11%	69.01%	9.53%	4.75%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	**0**	1.00E-03	1.00E-04
Learning Rate (Initialization)	45.77%	57.08%	-1.81%	-7.18%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	**1.00E-04**	1.00E-04
Learning Rate (Initialization)	1.00%	5.00%	-46.58%	-59.26%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	**1.00E-02**	1.00E-04
Learning Rate (Initialization)	30.50%	41.64%	-17.08%	-22.62%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	**1.00E-05**	1.00E-04
Weight Decay	48.28%	64.15%	0.70%	-0.11%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	1.00E-03	**1.00E-05**
Weight Decay	48.34%	62.05%	0.76%	-2.21%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	1.00E-03	**1.00E-03**
Weight Decay	45.36%	62.06%	-2.22%	-2.20%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	1.00E-03	**1.00E-06**

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay
Best Performing	57.11%	69.01%	0%	0%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0	1.00E-03	1.00E-04
Weight Decay	57.51%	68.94%	0.40%	-0.07%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0	1.00E-03	**1.00E-05**
Weight Decay	57.13%	69.50%	0.02%	0.49%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0	1.00E-03	**1.00E-06**
Conv/FC Layers	57.36%	69.35%	0.25%	0.34%	7,552,056	IN - 64 - 128 - 256	**IN - 1024 - 100**	**IN - 512 - 20**	ReLU	YES	0	1.00E-03	1.00E-06
Conv/FC Layers	57.18%	69.48%	0.07%	0.47%	13,957,688	IN - 64 - 128 - 256	**IN - 2048 - 100**	**IN - 1024 - 20**	ReLU	YES	0	1.00E-03	1.00E-06
Best Tweaked Model	57.36%	69.35%	0.25%	0.34%	7,552,056	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	0	1.00E-03	1.00E-06

The Highlighted value in the table is the one that was changed for that model.

Note: The model training statistics indicated that the current tweakeed model was severely overftting.

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay
Best Tweaked Model	57.36%	69.35%	57.36%	69.35%	7,552,056	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	0	1.00E-03	1.00E-06

Model Change 1: Adding Batchnorm to the FC Layers.

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay
Batchnorm on FC Layers	57.54%	67.69%	0.18%	-1.66%	7,555,128	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	0	1.00E-03	1.00E-06
Increase Dropout	58.17%	68.47%	0.81%	-0.88%	7,555,128	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	**0.3**	1.00E-03	1.00E-06
Increase Dropout	59.92%	69.42%	2.56%	0.07%	7,555,128	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	**0.5**	1.00E-03	1.00E-06
Increase Dropout	60.78%	71.10%	3.42%	1.75%	7,555,128	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	**0.7**	1.00E-03	1.00E-06
Increase Dropout	57.62%	69.44%	0.26%	0.09%	7,555,128	IN - 64 - 128 - 256	IN - 1024 - 100	IN - 512 - 20	ReLU	YES	**0.9**	1.00E-03	1.00E-06
Decrease FC Neurons	55.08%	67.54%	-2.28%	-1.81%	4,350,776	IN - 64 - 128 - 256	**IN - 512 - 100**	**IN - 256 - 20**	ReLU	YES	0.9	1.00E-03	1.00E-06
Decrease Dropout	60.97%	70.93%	3.61%	1.58%	4,350,776	IN - 64 - 128 - 256	**IN - 512 - 100**	**IN - 256 - 20**	ReLU	YES	0.7	1.00E-03	1.00E-06
Increase Dropout	58.64%	70.43%	1.28%	1.08%	4,350,776	IN - 64 - 128 - 256	**IN - 512 - 100**	**IN - 256 - 20**	ReLU	YES	0.8	1.00E-03	1.00E-06
Best	60.97%	70.93%	3.61%	1.58%	4,350,776	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.7	1.00E-03	1.00E-06
FC Batchnorm off	44.47%	59.70%	-12.89%	-9.65%	4,347,704	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.7	1.00E-03	1.00E-06
FC Batchnorm back on	59.20%	71.12%	1.84%	1.77%	4,350,776	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.7	1.00E-03	1.00E-06

Model Change 2: Dropout = .7 may indicates too many parameters so I will test the model size and depth

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay	
Best from change 1	59.20%	71.12%	0.00%	0.00%	4,350,776	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.7	1.00E-03	1.00E-06	
Decrease Parameters and Dropout	58.04%	68.41%	-1.16%	-2.71%	2,748,600	IN - 64 - 128 - 256	**IN - 256 - 100**	**IN - 128 - 20**	ReLU	YES	**0.35**	1.00E-03	1.00E-06	
Decrease Dropout	57.53%	67.26%	-1.67%	-3.86%	2,748,600	IN - 64 - 128 - 256	**IN - 256 - 100**	**IN - 128 - 20**	ReLU	YES	**0.15**	1.00E-03	1.00E-06	
Change FC Architecture	56.31%	68.63%	-2.89%	-2.49%	2,776,056	IN - 64 - 128 - 256	**IN - 256 - 128 - 100**	**IN - 128 - 64 - 20**	ReLU	YES	**0.15**	1.00E-03	1.00E-06	
Change FC Architecture	56.61%	68.90%	-2.59%	-2.22%	2,776,056	IN - 64 - 128 - 256	IN - 256 - 128 - 100	IN - 128 - 64 - 20	ReLU	YES	**0.3**	1.00E-03	1.00E-06	
Weight Decay Increase	57.02%	69.00%	-2.18%	-2.12%	2,776,056	IN - 64 - 128 - 256	IN - 256 - 128 - 100	IN - 128 - 64 - 20	ReLU	YES	0.3	1.00E-03	**1.00E-04**	
Weight Decay Increase	56.08%	68.83%	-3.12%	-2.29%	2,776,056	IN - 64 - 128 - 256	IN - 256 - 128 - 100	IN - 128 - 64 - 20	ReLU	YES	0.3	1.00E-03	**5.00E-04**	
Dropout Increase	56.28%	68.82%	-2.92%	-2.30%	2,776,056	IN - 64 - 128 - 256	IN - 256 - 128 - 100	IN - 128 - 64 - 20	ReLU	YES	**0.5**	1.00E-03	**1.00E-04**	
Change FC Architecture	56.26%	68.85%	-2.94%	-2.27%	2,776,056	IN - 64 - 128 - 256	IN - 256 - 128 - 100	IN - 128 - 64 - 20	ReLU	YES	0.3	1.00E-03	1.00E-04	**Note: This run I removed dropout after the 2nd FC layer that I added.**
Best	57.02%	69.00%	-2.18%	-2.12%	2,776,056	IN - 64 - 128 - 256	IN - 256 - 128 - 100	IN - 128 - 64 - 20	ReLU	YES	**0.3**	1.00E-03	1.00E-04	
Change FC Depth	57.80%	68.71%	-1.40%	-2.41%	2,748,600	IN - 64 - 128 - 256	**IN - 256 - 100**	**IN - 128 - 20**	ReLU	YES	**0.3**	1.00E-03	1.00E-04	
Increase Dropout	58.90%	70.45%	-0.30%	-0.67%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	**0.65**	1.00E-03	1.00E-04	
Moving on with	58.90%	70.45%	-0.30%	-0.67%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	**0.65**	1.00E-03	1.00E-04	

Model Change 3: LLM Recommendation added at the end of the script; Label Smoothing and loss weighting

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay	Label Smoothing	Loss Weighting (Fine, Coarse)
Moving on with	58.90%	70.45%	0.00%	0.00%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0**	1.0, 0.5
Label Smoothing	60.41%	70.76%	1.51%	0.31%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.15**	1.0, 0.5
Label Smoothing	60.65%	70.85%	1.75%	0.40%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.2**	1.0, 0.5
Label Smoothing	61.53%	71.24%	2.63%	0.79%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.25**	1.0, 0.5
Label Smoothing	61.15%	71.33%	2.25%	0.88%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.3**	1.0, 0.5
Label Smoothing	61.61%	71.13%	2.71%	0.68%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.4**	1.0, 0.5
Loss Weighting	60.48%	71.48%	1.58%	1.03%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.25**	**1.0, 1.0**
Loss Weighting	60.99%	70.77%	2.09%	0.32%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.25**	**1.0, 0.2**
Loss Weighting	60.45%	71.43%	1.55%	0.98%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.25**	**1.0, 0.75**
Best	61.53%	71.24%	2.63%	0.79%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	**0.4**	1.0, 0.5

Model Change 4: LLM Recomendation Skip Connections

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay	Label Smoothing
Best from model change 3	61.53%	71.24%	0.00%	0.00%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	0.25
With skip connections	57.97%	68.67%	-3.56%	-2.57%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	0.25

Model Change 5: LLM Recomendation Data Augmentation

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay	Label Smoothing	Data Augment
Best from model change 3	61.53%	71.24%	0.00%	0.00%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	0.4	NO
With Data Augmentation	65.47%	76.25%	3.94%	5.01%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	1.00E-03	1.00E-04	0.4	YES
Increase Learning Rate Initialization	66.26%	77.62%	4.73%	6.38%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	**1.00E-02**	1.00E-04	0.4	YES
Increase Learning Rate Initialization	22.74%	50.49%	-38.79%	-20.75%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.65	**5.00E-02**	1.00E-04	0.4	YES
Decrease Dropout	67.51%	77.75%	5.98%	6.51%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	**0.45**	**1.00E-02**	1.00E-04	0.4	YES
Decrease Dropout	67.91%	77.61%	6.38%	6.37%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	**0.25**	1.00E-02	1.00E-04	0.4	YES

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay	Label Smoothing	Data Augment
Original Model	47.58%	64.26%	0.00%	0.00%	4,349,240	IN - 64 - 128 - 256	IN - 512 - 100	IN - 256 - 20	ReLU	YES	0.4	1.00E-03	1.00E-04	0	NO
Final Model	67.91%	77.61%	20.33%	13.35%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.25	1.00E-02	1.00E-04	0.25	YES

Ablation Study on the Final Model

	Top-1 Acc (Fine)	Top-1 Acc (Coarse)	Difference (Fine)	Difference (Coarse)	Paramters	Conv Layers	FC Layer (Fine)	FC Layer (Coarse)	Activation	Batchnorm	Dropout	Learning Rate (Initialization)	Weight Decay	Label Smoothing	Data Augment
Final Model	67.91%	77.61%	0.00%	0.00%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.25	1.00E-02	1.00E-04	0.4	YES
Remove Batchnorm	1.00%	5.00%	-66.91%	-72.61%	2,746,040	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	**NO**	0.25	1.00E-02	1.00E-04	0.4	YES
Remove Batchnorm from FC	28.84%	46.29%	-39.07%	-31.32%	2,747,832	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	**NO**	0.25	1.00E-02	1.00E-04	0.4	YES
Remove Dropout	64.81%	76.54%	-3.10%	-1.07%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	**0**	1.00E-02	1.00E-04	0.4	YES
Remove Weight Decay	66.92%	77.11%	-0.99%	-0.50%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.25	1.00E-02	**0.00E+00**	0.4	YES
Remove Label Smoothing	64.94%	76.04%	-2.97%	-1.57%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.25	1.00E-02	1.00E-04	**0**	YES
Remove Data Augmentation	57.96%	68.67%	-9.95%	-8.94%	2,748,600	IN - 64 - 128 - 256	IN - 256 - 100	IN - 128 - 20	ReLU	YES	0.25	1.00E-02	1.00E-04	0.4	**NO**